

Rebuild from Inorder & Postorder

Binary Trees, Construction & Reconstruction · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★★ Advanced

Problem

Reconstruct a binary tree from its inorder and postorder traversals (all values distinct), then report the tree's preorder traversal and its height.

Implement the reconstruction, the preorder walk and the height from scratch.

What to Compute

- Preorder: the preorder traversal (node, left, right) of the reconstructed tree.
- Height: the height of the tree in edges (a single node is 0).

Input / Output Format

Input: Line 1: the inorder sequence. Line 2: the postorder sequence.

Output: Two lines: Preorder and Height.

Examples

Example 1

Input: inorder = 9 3 15 20 7, postorder = 9 15 7 20 3

Output:

Preorder: 3 9 20 15 7
Height: 2

Explanation: The last postorder value 3 is the root; it splits inorder into left [9] and right [15 20 7], yielding a tree of height 2 whose preorder is 3 9 20 15 7.

Example 2

Input: inorder = 1 2 3, postorder = 1 3 2

Output:

Preorder: 2 1 3
Height: 1

Explanation: Postorder's last value 2 is the root with children 1 and 3; preorder is 2 1 3 and the height is 1.

Constraints

- All values are distinct.
- Inorder lists left subtree, node, right subtree; postorder lists left, right, node.
- Height is in edges (single node = 0).

Sample Run

Sample Input

```
9 3 15 20 7
9 15 7 20 3
```

Sample Output

```
Preorder: 3 9 20 15 7
Height: 2
```